Supervised Classification

Supervised Classification

- The process of using samples of known classes to classify the remaining unknown pixels to these classes within the image
- Supervised classification exploits prior knowledge about the data to identify which of the predefined informational classes closely resembles a classification unit
- This prior knowledge typically is in the form of training samples

Supervised Classification

- **Analyst first selects training samples/ signatures for all representative classes (classes of interest)**
- **These training samples are a collection of representative data points for which the class labels are known to the analyst**
- **Statistical characteristics (based on spectral information) of these training samples are used to train the classifier, to identify and match the class labels of the unknown data points**

Supervised Classification

- As the classification process is based on the training data, they must be homogeneous and represent the classes as closely as possible
- Pixels that are at the boundaries of the different information classes, (pixels with mixed spectral characteristics) should not be selected as training pixels to avoid between-class confusions
- The training samples for a class should represent within-class spectral variations and should not include pixels with obvious mixtures

Supervised Classification

- The training samples should be well spread over the entire image to adequately represent the information classes
- An image with high within-class spectral variability and low between-class spectral similarity makes it hard to select true representative class samples and to achieve satisfactory classification results
- Training pixels should be sufficient in numbers , and the general rule is to collect at least 10xN to 100xN pixels for each class, where N is the number of bands of the image being used

Supervised Classification

- **Brightness value of each pixel of the training data of the desired classes are used to calculate the mean, standard deviation, variance-covariance and correlation matrices**
- **These measures represent the spectral characteristics of classes, and required by the algorithms to assign appropriate class to the remaining unclassified image data**

Optimum Bands Selection for Image Classification: Feature Selection

- After getting training statistics, a judgment is often made to determine the bands that are most effective in discriminating each class from all others
- This process is commonly called ***feature selection***
- The goal is to exclude the bands from the analysis that provide redundant spectral information
- This minimizes the cost of the digital image classification process

SUPERVISED CLASSIFICATION

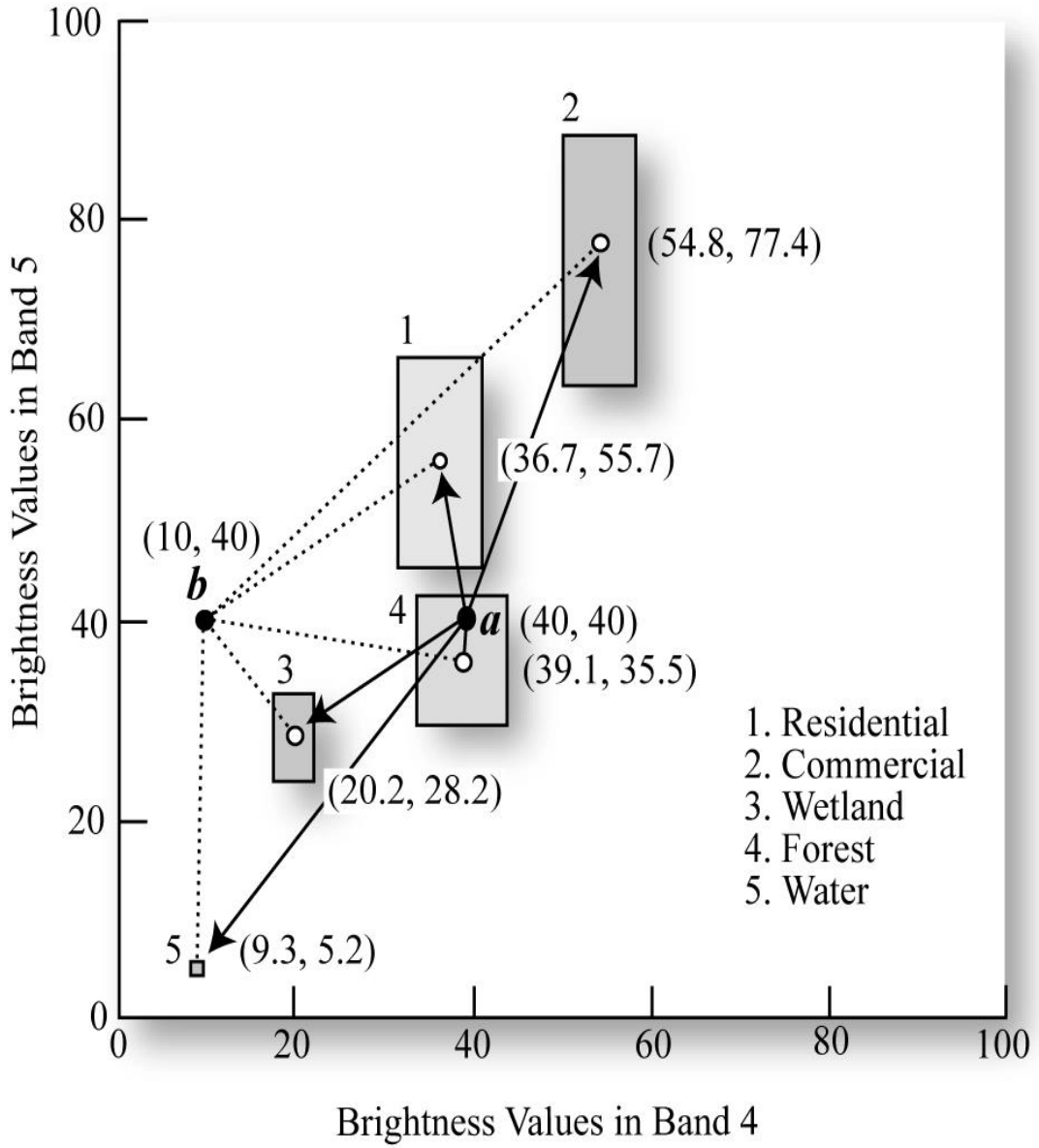
- **Training stage**
  - Analyst need to have significant prior knowledge of the area and land covers
  - Collection of appropriate Training sample or training signature
- **Band Selection**
- **Classification Stage** : categorization of each pixel to landscape class it most closely resemble
- **Accuracy Assessment**
- **Output stage**: the results may be used to:
  - Generate thematic maps
  - Tabulate scene/classification statistics
  - Use of the derived information for further analysis in GIS

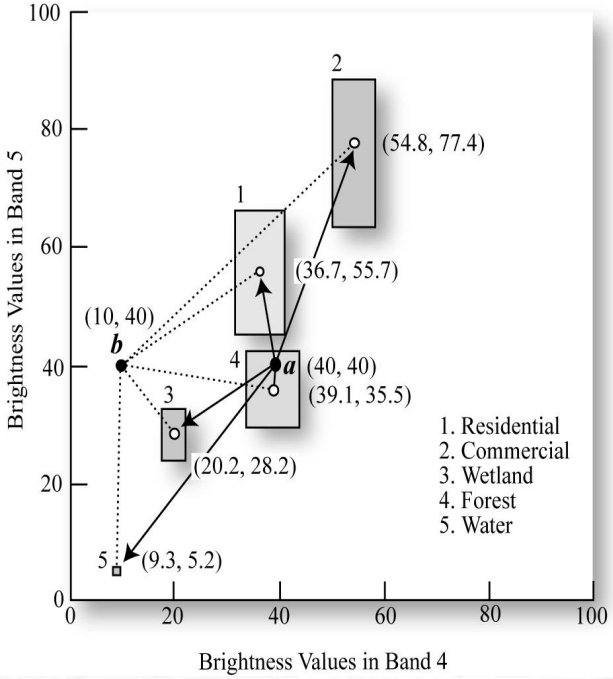
Minimum Distance to Means Classification Algorithm

To assign unclassified pixels to their nearest classes

- **The minimum distance to means decision rule is computationally simple and commonly used**
- **It requires that the user provide the mean vectors for each class in each band μ_{ck} from the training data**
- **To perform a minimum distance classification, a program must calculate the distance to each mean vector μ_{ck} from each unknown pixel (BV_{ijk})**

Minimum Distance to Means Classification Algorithm

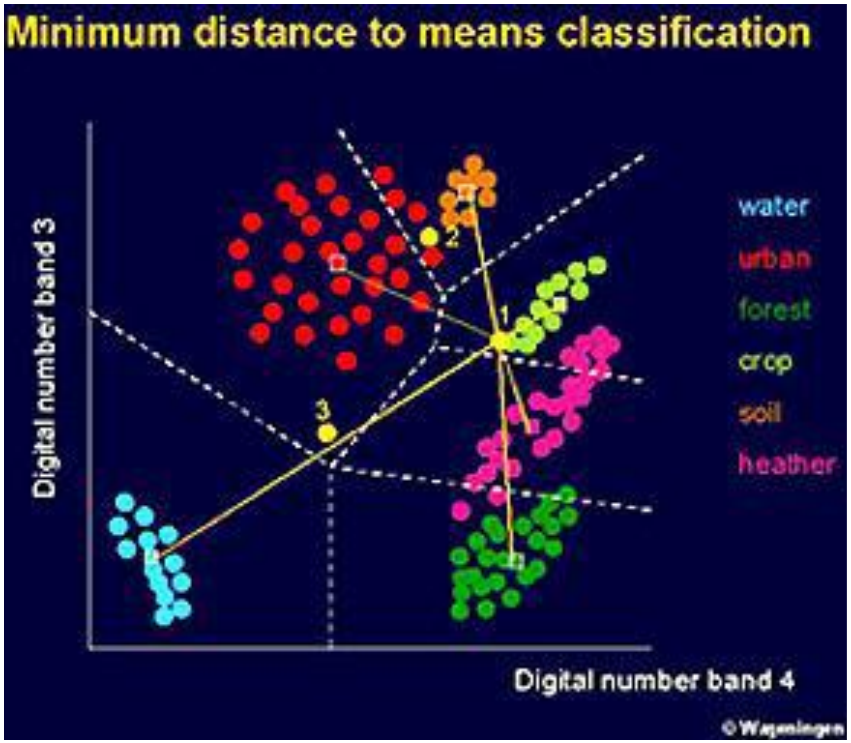




	Class	Distance from pixel *a* (40, 40) to the mean of each class	Distance from pixel *b* (10, 40) to the mean of each class
1.	Residential	$\sqrt{(40-36.7)^2+(40-55.7)^2} = 16.04$	$\sqrt{(10-36.7)^2+(40-55.7)^2} = 30.97$
2.	Commercial	$\sqrt{(40-54.8)^2+(40-77.4)^2} = 40.22$	$\sqrt{(10-54.8)^2+(40-77.4)^2} = 58.35$
3.	Wetland	$\sqrt{(40-20.2)^2+(40-28.2)^2} = 23.04$	$\sqrt{(10-20.2)^2+(40-28.2)^2} = 15.75$ Assign pixel *b* to this class; it has the minimum distance
4.	Forest	$\sqrt{(40-39.1)^2+(40-35.5)^2} = 4.59$ Assign pixel *a* to this class; it has the minimum distance	$\sqrt{(10-39.1)^2+(40-35.5)^2} = 29.45$
5.	Water	$\sqrt{(40-9.3)^2+(40-5.2)^2} = 46.4$	$\sqrt{(10-9.3)^2+(40-5.2)^2} = 34.8$

MINIMUM DISTANCE TO MEANS

- Point 1, an unknown, the shortest straight-line distance to the several means is to the class "----------Soil----------". Point 1, then, is assigned to ----------Soil----------✔Crops----------
- Point 2 is slightly closer to the "soil" but lies at the edge of the "urban" spread. Here, the classification seems ambiguous.
  - By the minimum distance rule, it would go to "soil" but this may be erroneous ("urban" would have been a greater likelihood).
- Point 3 is not near any of the class DN clusters, but is about equidistance between "urban", "water", "forest", and "heather". If one plays the odds, "urban" may be closer to 3; but this situation indicates how misclassification might occur.



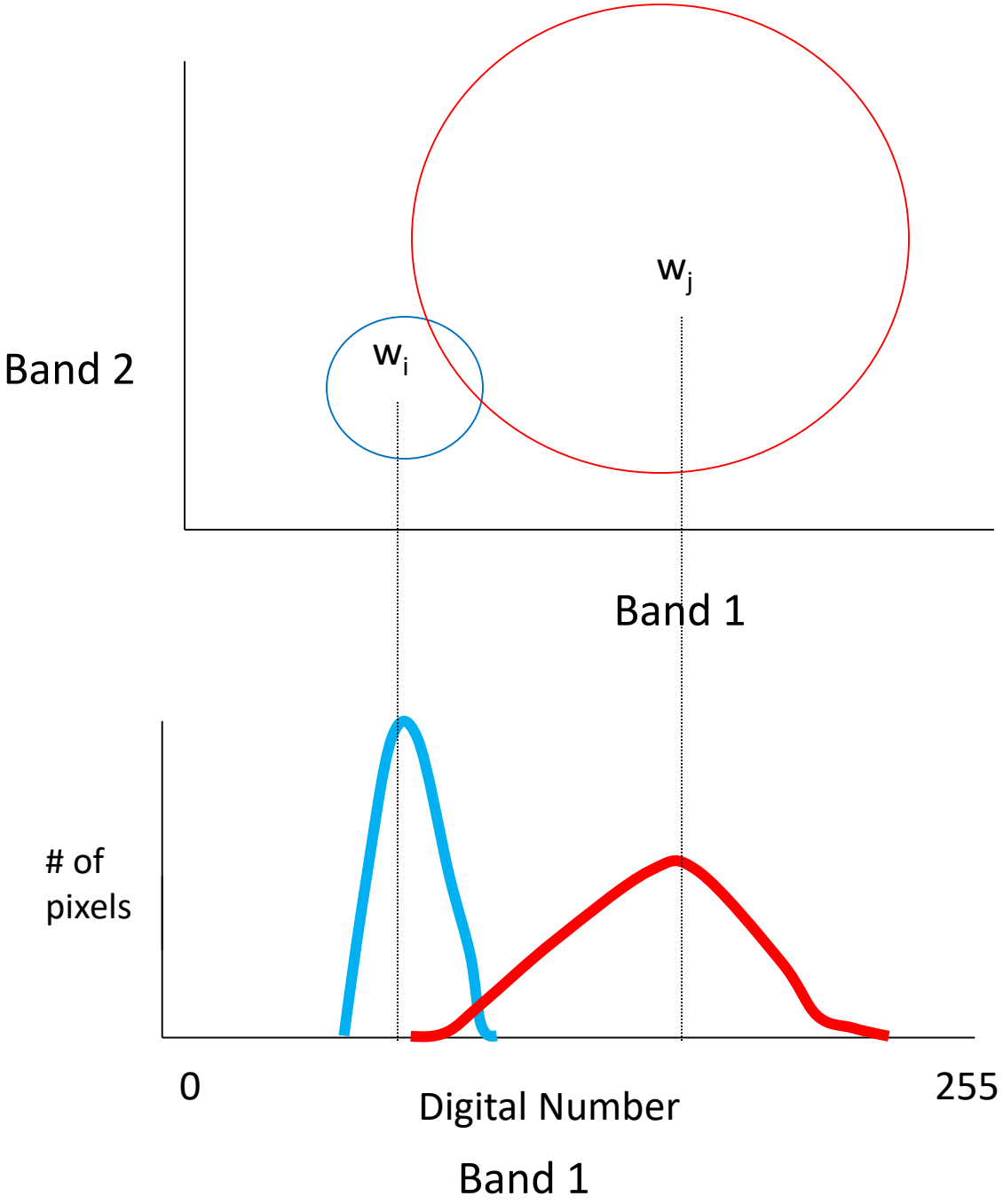
Classification Algorithm

- Several widely adopted classification algorithms exits
- The most widely adopted parametric classification algorithms is the Maximum likelihood
- These algorithms require that
  - The number of classes be specified in advance, and
  - That certain statistical characteristics of each class to which the pixels are to be allocated are known, these estimates are derived from the training samples

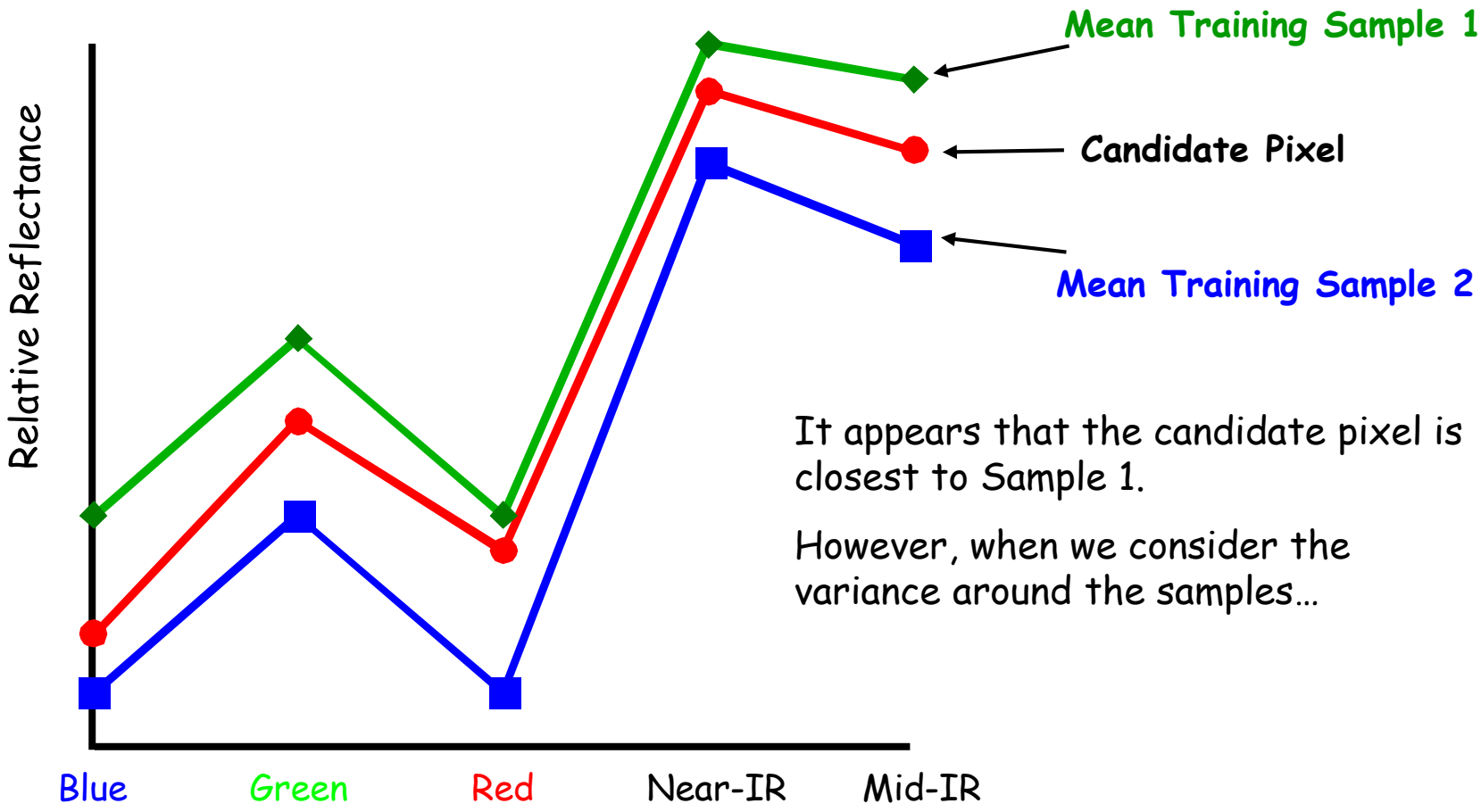
Maximum Likelihood Classification Algorithm

- **The maximum likelihood decision rule is based on probability**
- **The probability of a pixel belonging to each of a predefined set of m classes is calculated, and the pixel is then assigned to the class for which the probability is the highest**
- **The mean, variance and co-variance of each training class is needed to compute the probability function**

Class overlap



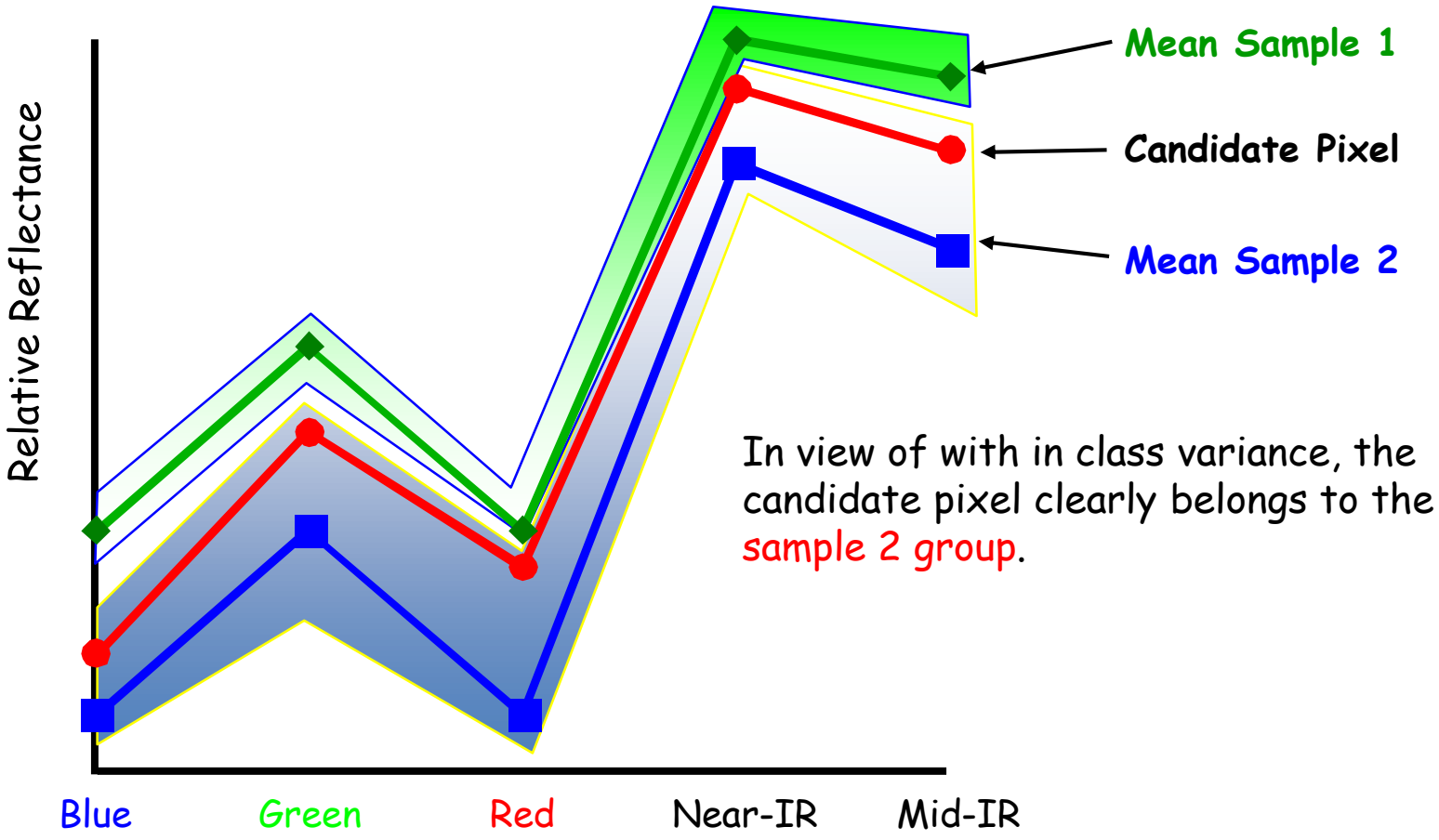
Maximum Likelihood Classifier



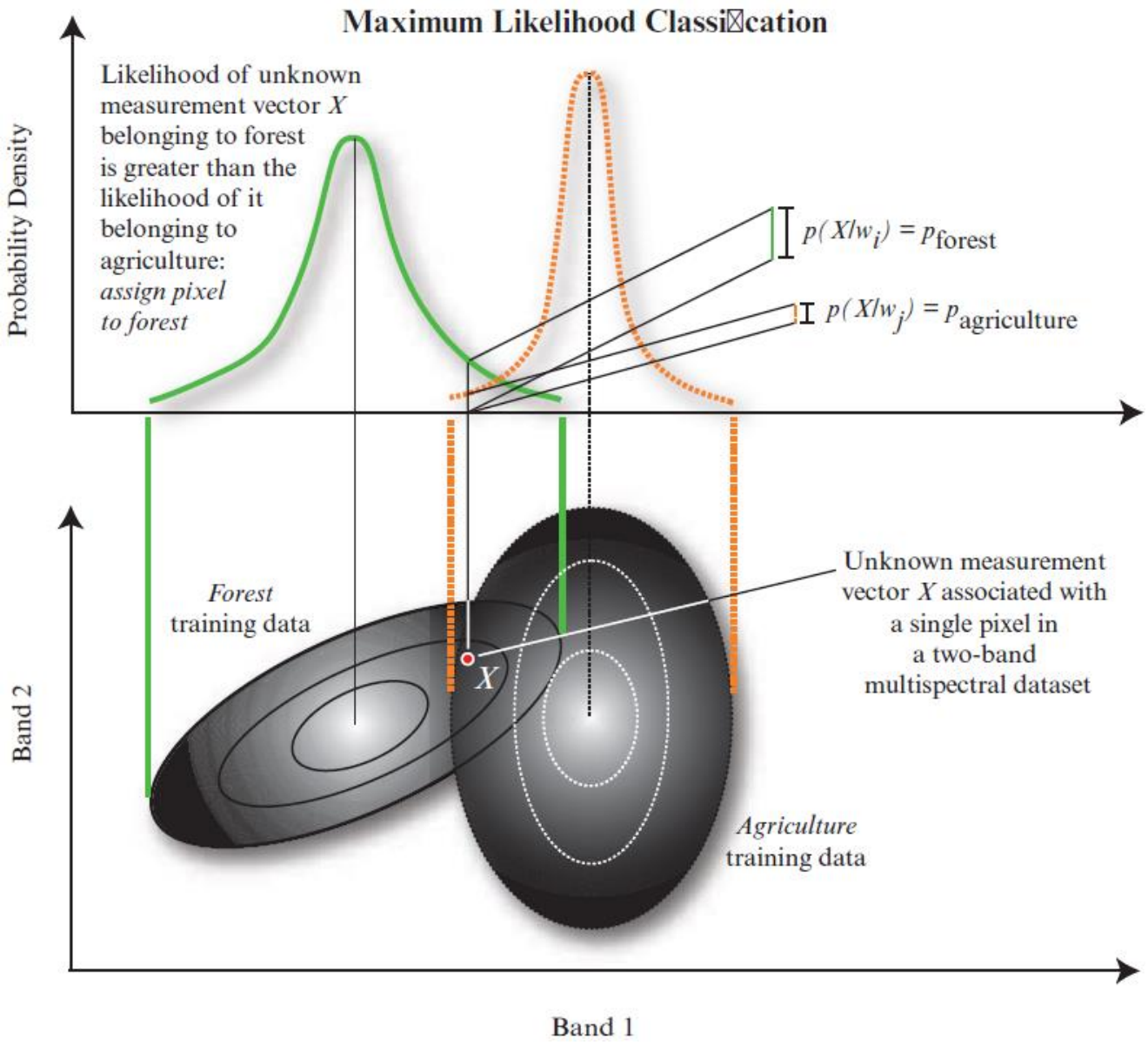
It appears that the candidate pixel is closest to Sample 1.

However, when we consider the variance around the samples…

Maximum Likelihood Classifier



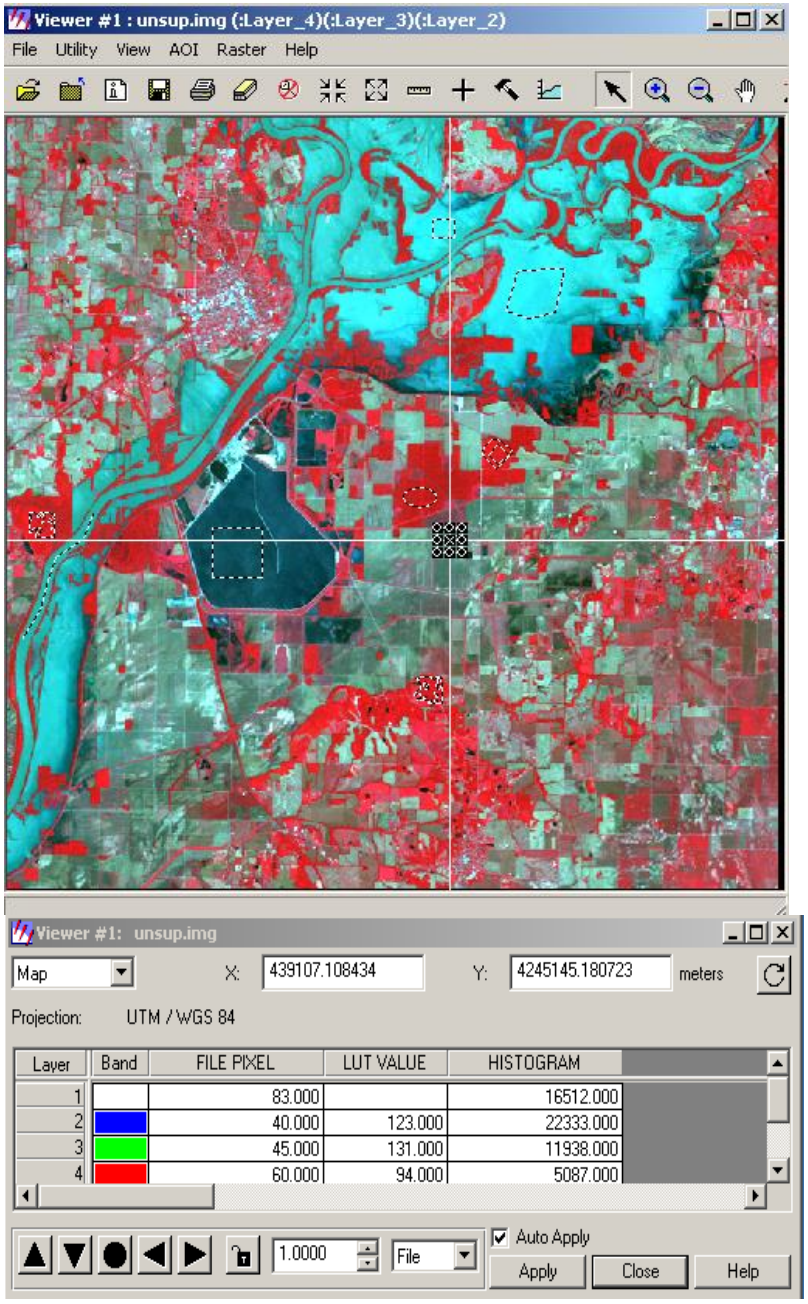
In view of with in class variance, the candidate pixel clearly belongs to the sample 2 group.

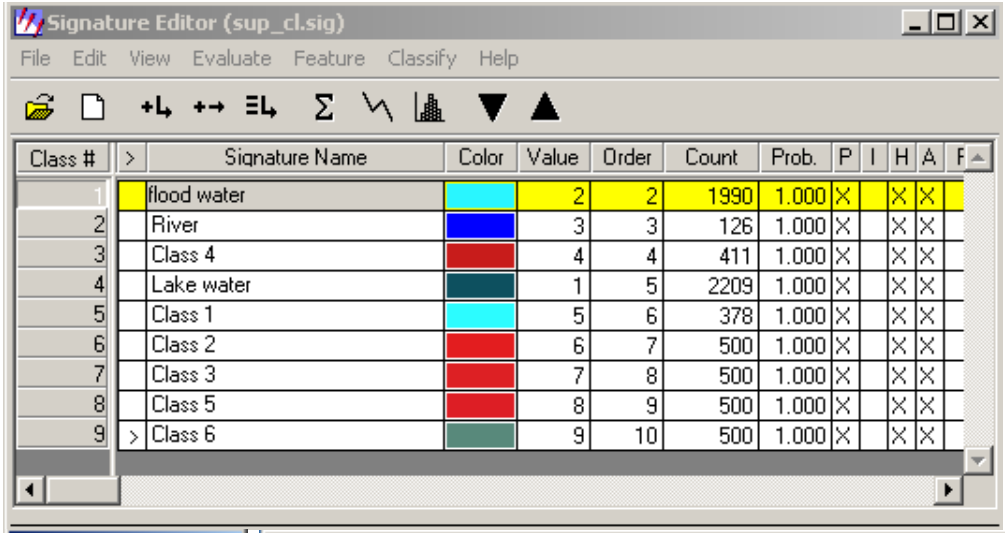


What happens when the probability density functions of two or more training classes overlap in feature space? For example, consider two hypothetical normally distributed probability density functions associated with forest and agriculture training data measured in bands 1 and 2. In this case, pixel *X* would be assigned to forest because the probability density of unknown measurement vector *X* is greater for forest than for agriculture.

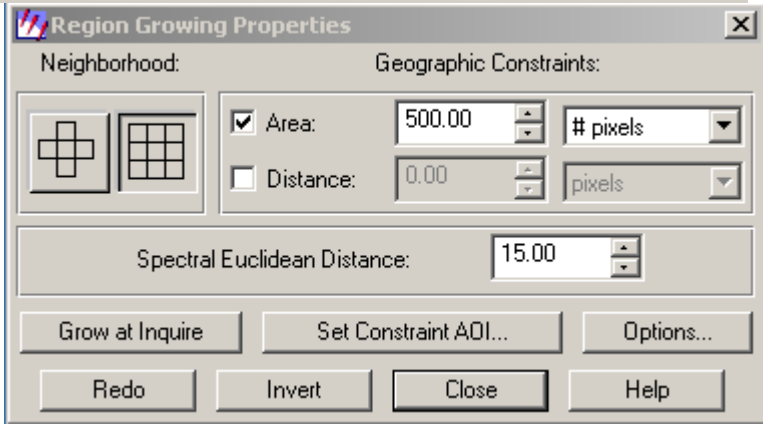
Summary -Training Samples Guidelines

- **Minimum number of samples = number of parameters to be estimated:**
- **Practical minimum = 10 to 100 x no. bands.**
- **Distribute training pixels for a spectral class across the image to better represent the variation of the class in the image.**
- **Do not include pixels in training areas that are adjacent to borders of spectral classes (like field boundaries).**
- **Do not include pixels for groups containing obvious mixtures (e.g. deep water and shallow water)**



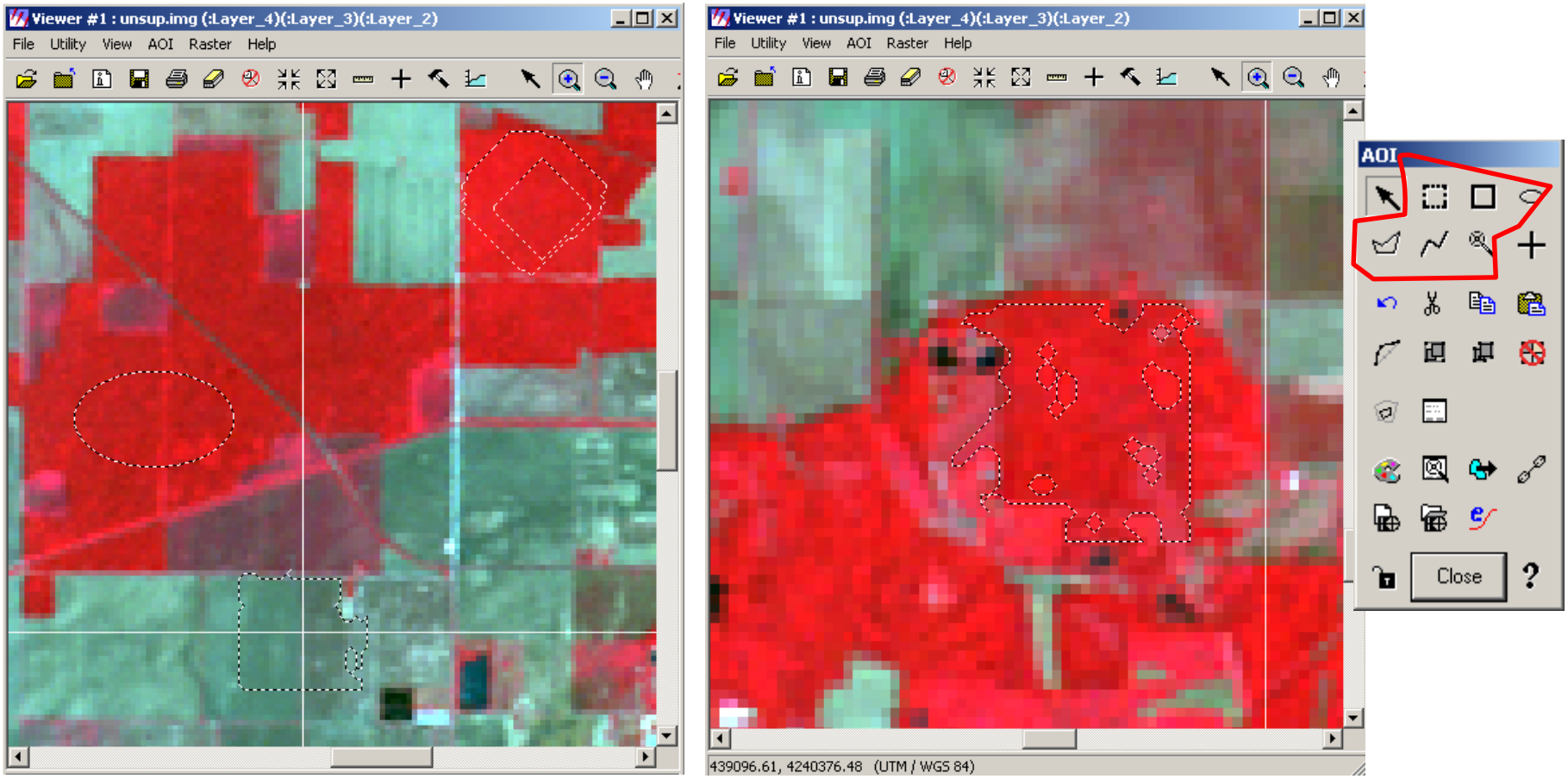






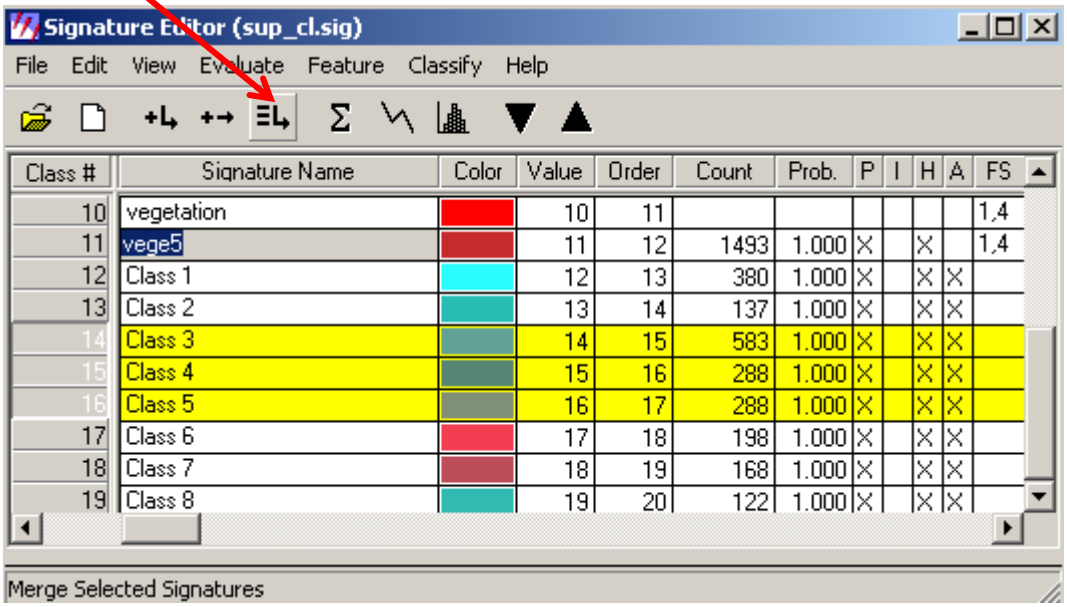
Different techniques for signatures selection

Signature Styles



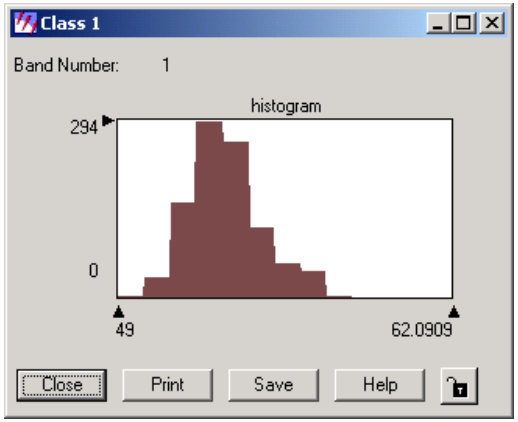
Merge Signature of Same Class

Select the classes need to be merged> shift click Class #
Press “ merge selected signature button
New class of merged signatures is added at the end
Delete the previous selected on and assign name to the merged signatures

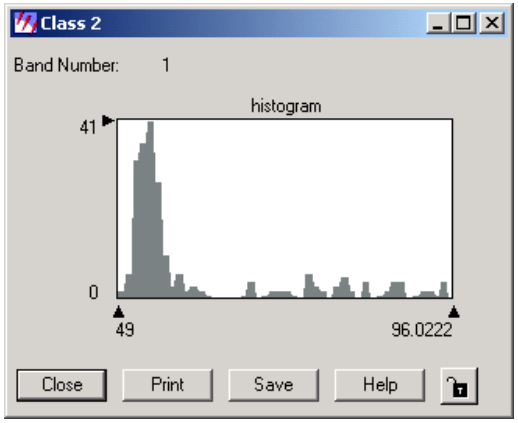


Signatures Accuracy

Which one will be a good signature?

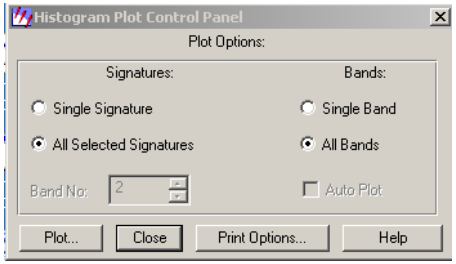


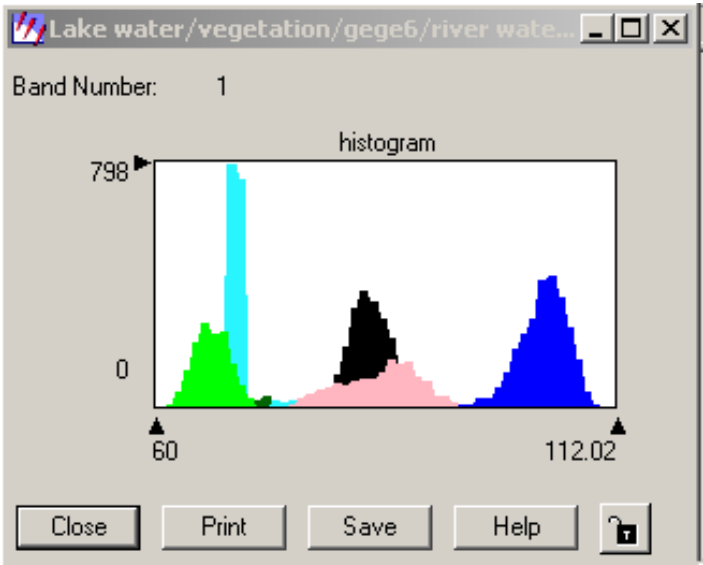
“good”

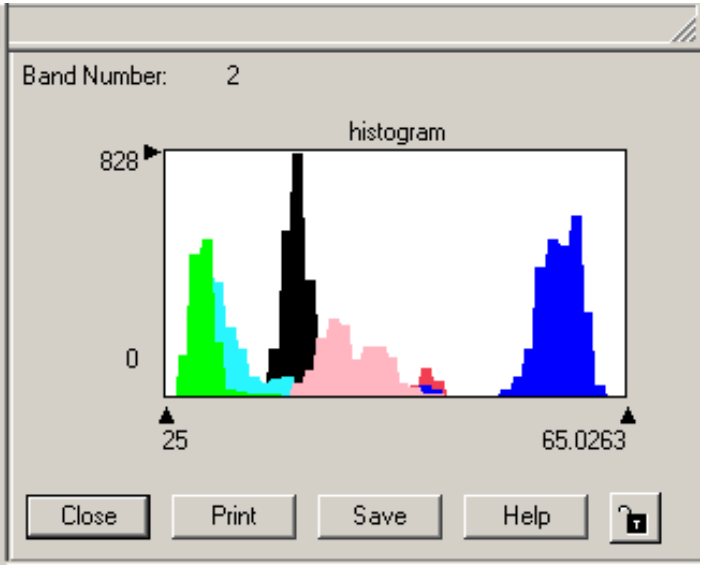


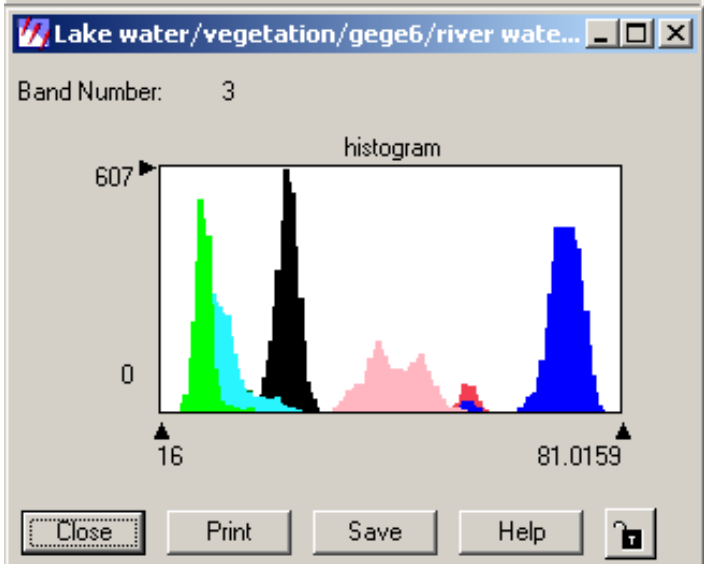
“bad”

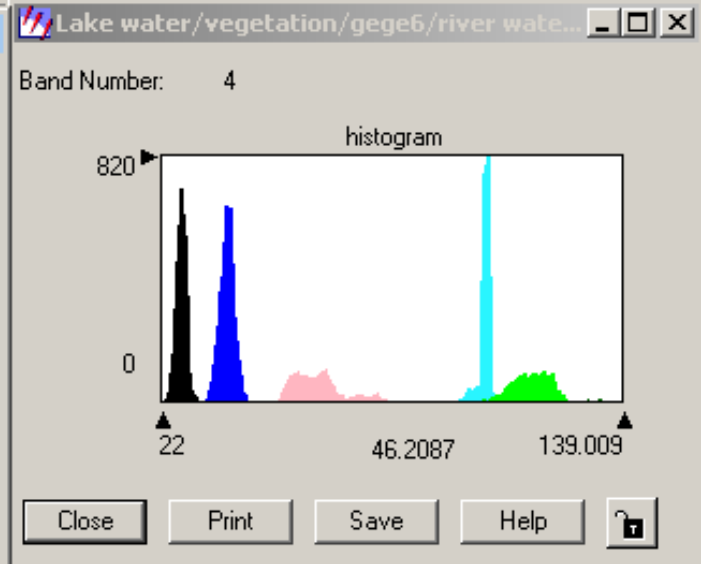
Signatures Accuracy-Histogram



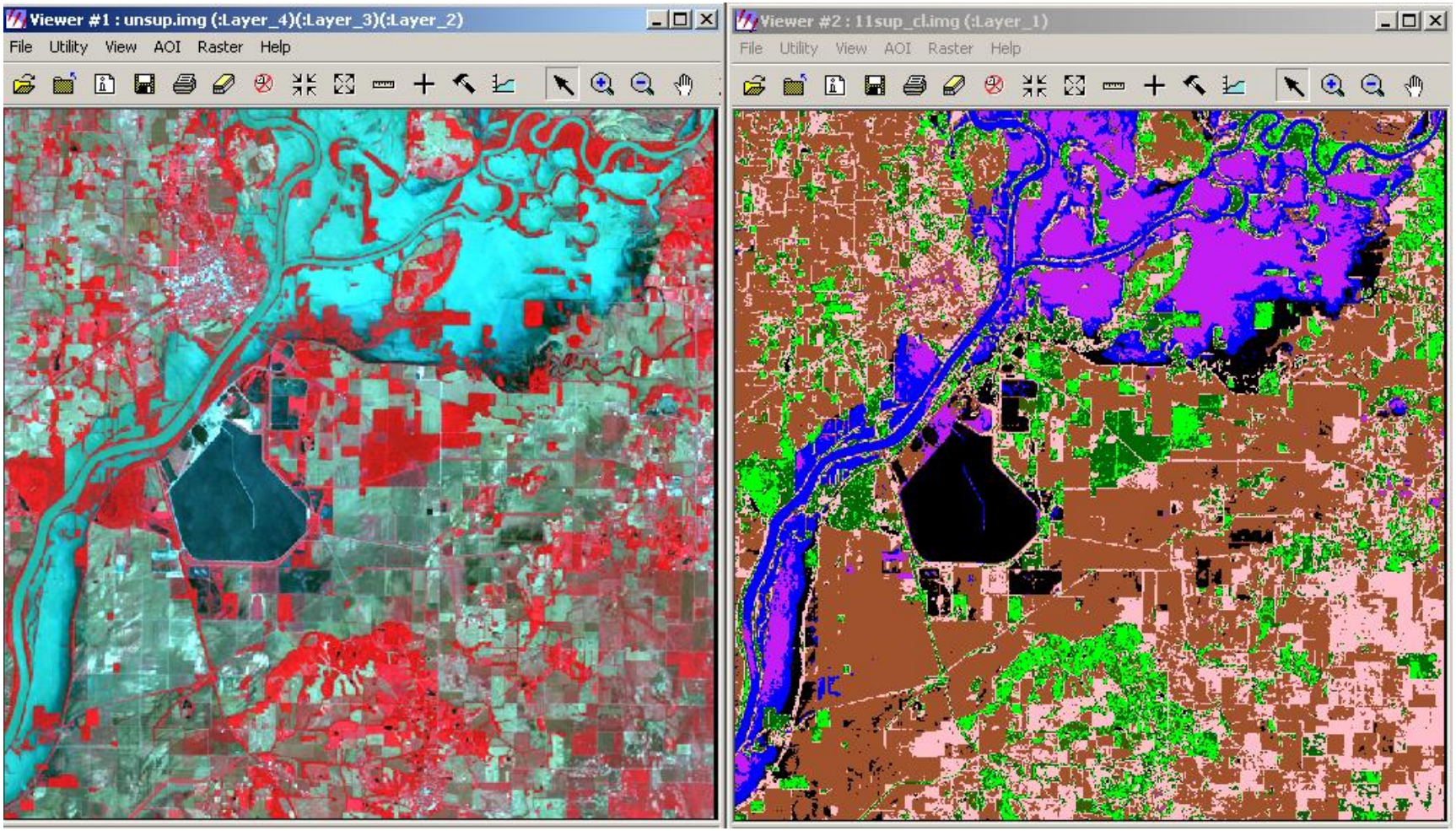




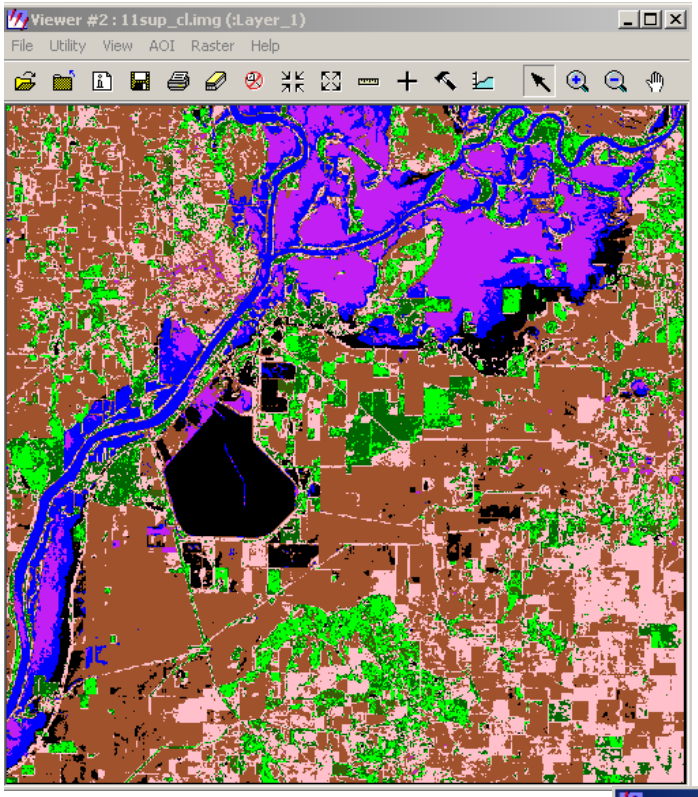


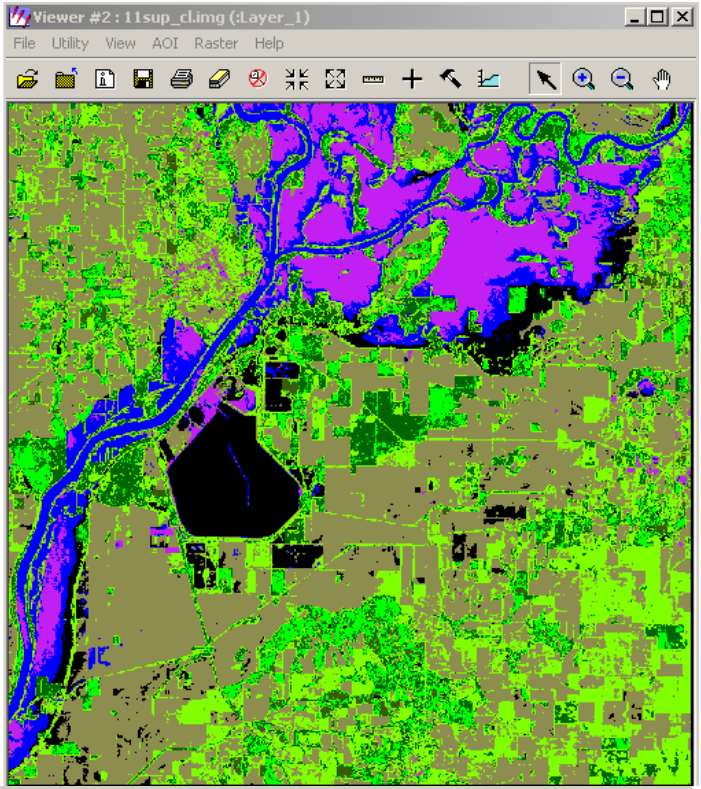


Original and classified image



Class Thematic Map





- Assign color to classes
- Add area column

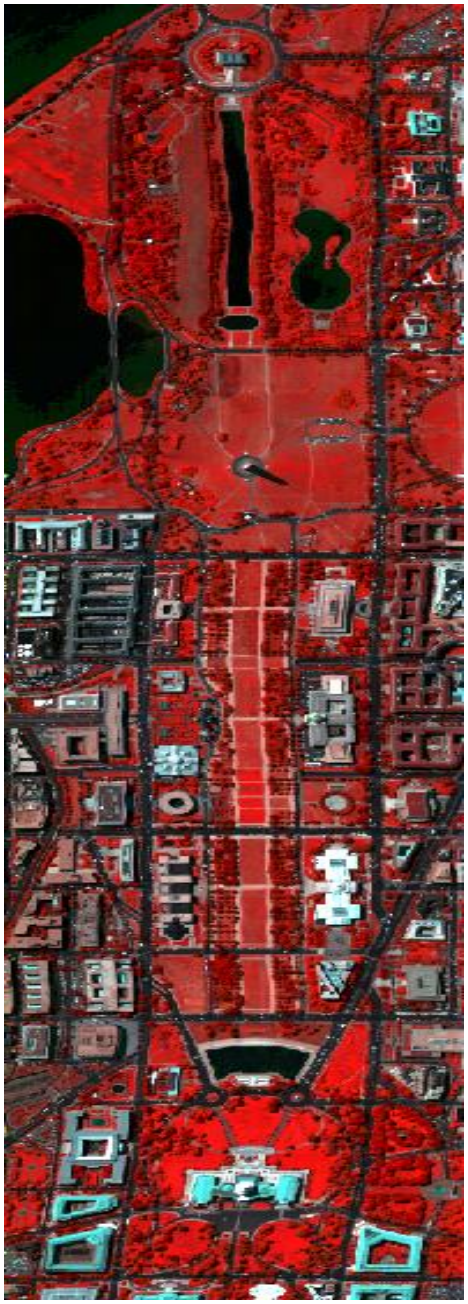
Raster Attribute Editor - 11sup_cl.img(:Layer_1)

File Edit Help

Layer Number: 1

Row	Class Names	Color	Area	Opacity	Red
18	openarea		18095.7	1	0.551
15	vege1		9943.38	1	0.498
12	vege11		4127.94	1	
9	flood water		3519.27	1	0.755
5	river water		3496.59	1	
6	vegetation		3160.44	1	
4	lake water		3019.14	1	
10	vege6		905.58	1	0.229
3			0	0	

Washington DC Mall data set.



MINIMUM DISTANCE TO MEAN METHOD

Key

- Roof
- Street
- Path
- Grass
- Trees
- Water
- Shadow

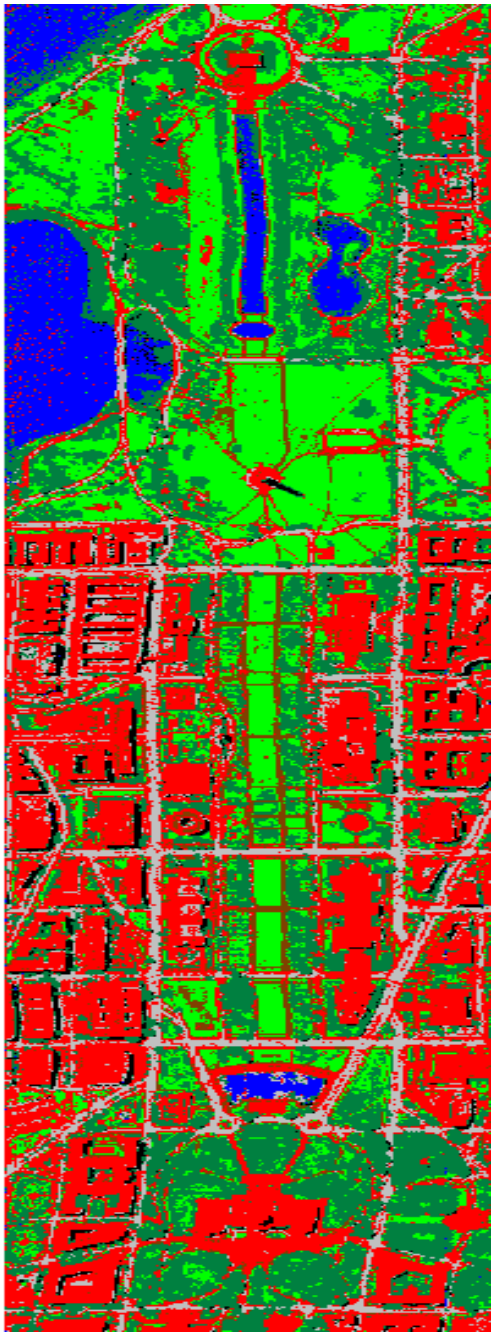


MAXIMUM LIKELYHOOD METHOD

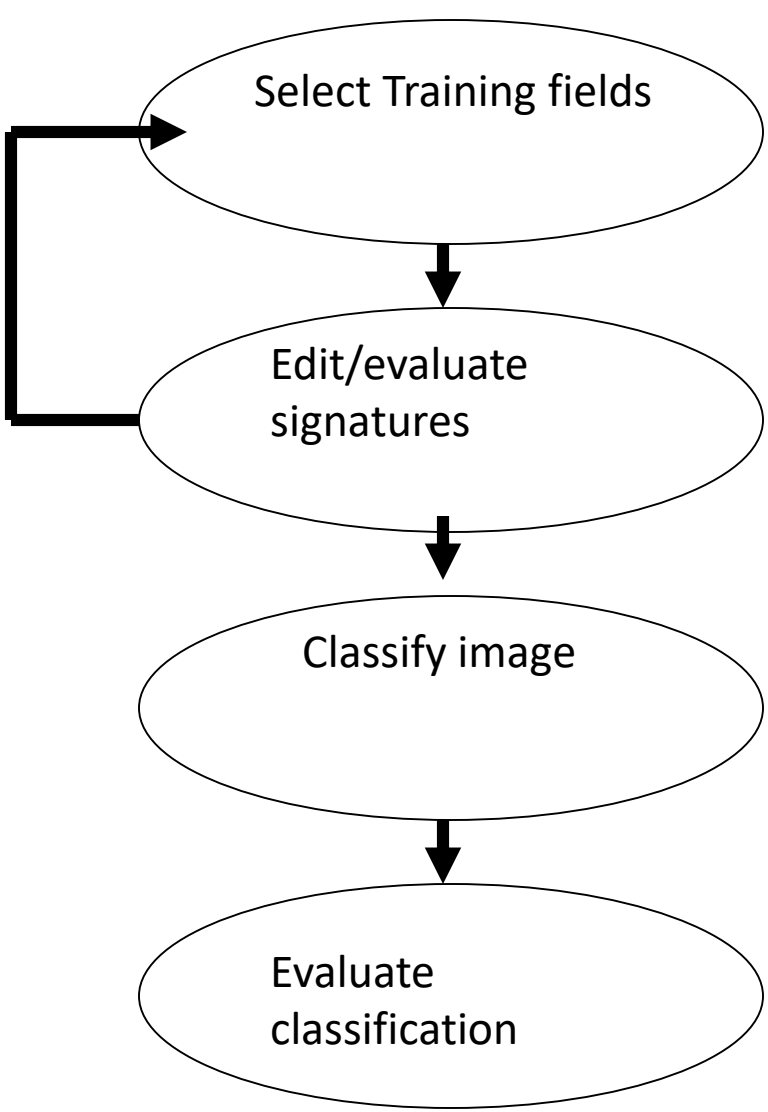


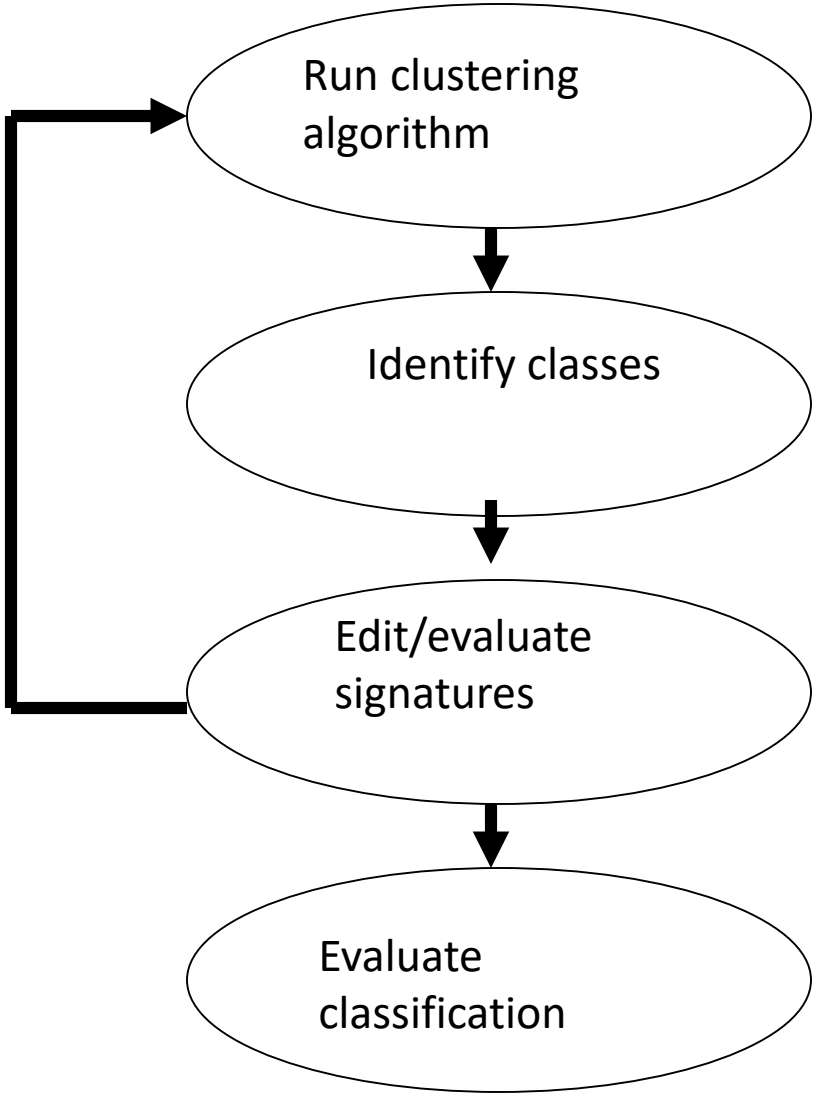
Key

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Supervised vs. Unsupervised





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